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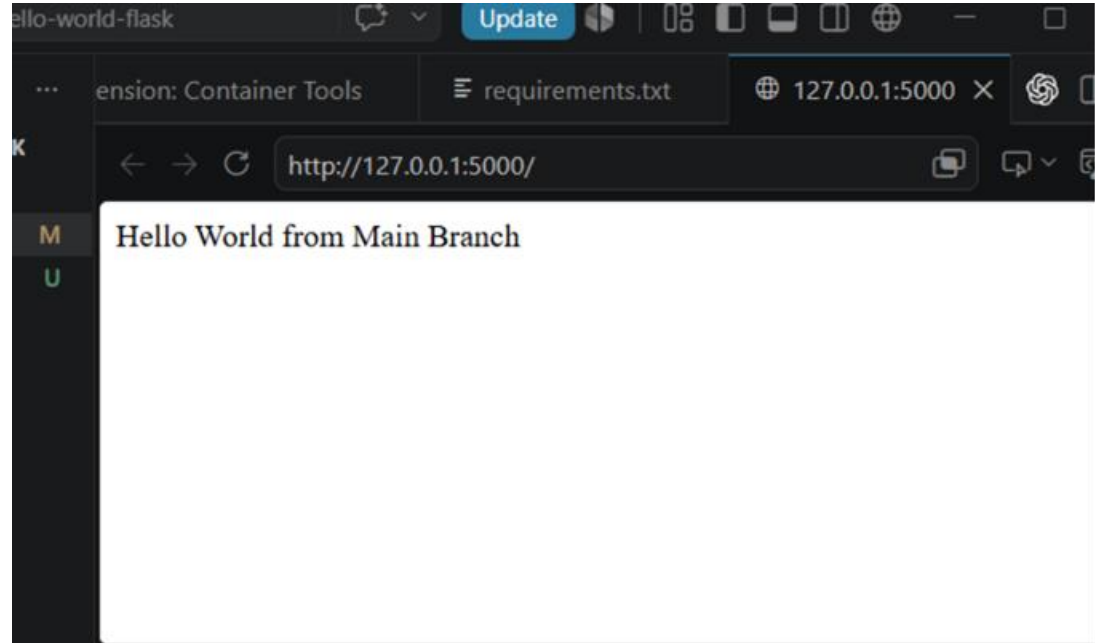
3.Basic Containerization (Docker) & Jenkins Freestyle Project (3 Marks)

- **Task 3.1:** Create a Dockerfile for your Python Flask "Hello World" application. The Dockerfile should build a minimal image that runs the application on port 5000. Build the Docker image locally and verify it runs correctly. (1.5 Marks)

```
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES

```

```
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> docker build -t hello-world-flask .
=> [3/4] COPY . .
=> [4/4] RUN pip install Flask
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:0e431f8304a91254accb6ae8aaeab1f422c9613ddb7c92188a8d83049906d9d
=> => exporting config sha256:9c8e010b2ad974d0591f889af664c3441c1e8c23a44ab1d264237683191803b5
=> => exporting attestation manifest sha256:9b4163ecf88dc6e9e35870e15c41fbfd988dceabf12d93f040b2fa1f012eb16a
=> => exporting manifest list sha256:3a0f252af6aaff574ab8565d2bae788246efbccbb8407f2b416dc4f3c3987758
=> => naming to docker.io/library/hello-world-flask:latest
=> => unpacking to docker.io/library/hello-world-flask:latest
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> docker images
IMAGE                                ID                                  DISK USAGE  CONTENT SIZE  EXTRA
hello-world-flask:latest             3a0f252af6aa                       200MB       49MB
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask>
```



The screenshot shows a web browser window with the address bar displaying 'http://127.0.0.1:5000/'. The page content displays 'Hello World from Main Branch'. The browser's developer tools are open, showing the 'Console' tab with the message 'Hello World from Main Branch'.

- **Task 3.2:** Set up a Jenkins Freestyle project. Configure it to pull your Git repository (from Task 1.1) and execute a build step that lists the contents of the workspace. Trigger a build and provide a screenshot of the successful build output. (1.5 Marks)

Getting Started



✓ Folders	✓ OWASP Markup Formatter	✓ Build Timeout	✓ Credentials Binding
✓ Timestamper	✓ Workspace Cleanup	✓ Ant	✓ Gradle
✓ Pipeline	🔄 GitHub Branch Source	🔄 Pipeline: GitHub Groovy Libraries	🔄 Pipeline Graph View
✓ Git	🔄 SSH Build Agents	🔄 Matrix Authorization Strategy	🔄 LDAP
🔄 Email Extension	✓ Mailer	🔄 Dark Theme	

```
! PIPELINE: Stage tags irrelevant
** Pipeline: Input Step
** Pipeline: Basic Steps
** Pipeline: Declarative
Pipeline
** Java JSON Web Token (JWT)
** GitHub API
** Mina SSHD API :: Common
** Mina SSHD API :: Core
** Gson API
** Git client
Git+
```

```
Started by user Anushka Desai
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\HelloWorld-Flask
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/AnushkaDesai13/hello-world-flask.git
> git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\HelloWorld-Flask # timeout=10
Fetching upstream changes from https://github.com/AnushkaDesai13/hello-world-flask.git
> git.exe --version # timeout=10
> git --version # 'git version 2.49.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/AnushkaDesai13/hello-world-flask.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe config remote.origin.url https://github.com/AnushkaDesai13/hello-world-flask.git # timeout=10
> git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 20b2c7d01e1b55bf37751c7139955ee4fa94f2ba
(refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 20b2c7d01e1b55bf37751c7139955ee4fa94f2ba # timeout=10
```